

Recall that E and F be events of an experiment

Bayes's Formula states that

$$P(E) = P(E/F) P(F)$$

$$+ P(E/F^c) P(F^c) -$$

Problem 1 An insurance company believes that people can be divided into two classes:

those who are accident prone and those who are not.

The Company's statistics show that an accident prone person

will have an accident at some time within a fixed 1-year period with probability 0.4, whereas the probability decreases to 0.2 for a person in non-accident prone.

If 30% of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing policy?

Solution:-

Let A_1 be the event that

the policy holder will have an accident within a year of purchasing policy.

Let A be the event that the policy holder is accident prone.

By Bayes's formula

$$p(A_1) = p(A) p(A_1/A) + p(A^c) p(A_1/A^c)$$

$$= (0.3)(0.4) + (0.7)(0.2)$$

$$= 0.12 + 0.14$$

$$= \underline{\underline{0.26.}}$$

Part II of problem ①

Suppose that a new policy holder has an accident within a year of purchasing a policy. What is the probability that he (or) she is accident prone?

Solⁿ:- We need to compute the probability that a new policy holder is accident prone given that he (or) she has an accident within a year of purchasing policy

$$P(A/A_1)$$

$$= \frac{P(A \cap A_1)}{P(A_1)}$$

Here $P(A \cap A_1)$ can be written as

$$P(A \cap A_1) = P(A) P(A_1/A)$$

Therefore,

$$\begin{aligned} P(A/A_1) &= \frac{P(A) P(A_1/A)}{P(A_1)} \\ &= \frac{(0.3)(0.4)}{0.26} \\ &= \frac{6}{13} \end{aligned}$$

Problem (2)

Consider a game played with a deck of 52 cards. The cards are shuffled and then turned over one at a time. At any time, the

player can guess that the next card turned over will be the ace of spades. If it is, then the player wins.

In addition, the player is said to win if the ace of spades has not yet appeared when only one card remains and no guess has yet been made.

What is the probability of winning?

Solution: Let G be the event that first card is guessed

Let $P(G) = p$. Then by

Bayes's theorem

$$\begin{aligned} P(\text{win}) &= P(\text{win}/G) P(G) \\ &\quad + P(\text{win}/G^c) P(G^c) \\ &= \frac{1}{52} p + \frac{1}{52} (1-p) \\ &= \underline{\underline{\frac{1}{52}}} \end{aligned}$$

Problem ③

A laboratory blood test is 95% effective in detecting a certain disease when it is, in fact, present. However, the test also yields a

"false positive" result for 1% of healthy persons tested. If 0.5% of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

Solution:

Let D be the event that the person tested has the disease and E be the event that the test result is positive.

The desired probability,

$$P(D/E)$$

$$= \frac{P(D \cap E)}{P(E)}$$

$$= \frac{P(D) P(E/D)}{P(D) P(E/D) + P(D^c) P(E/D^c)}$$

$$= \frac{(0.005)(0.95)}{(0.005)(0.95) + (0.995)(0.01)}$$

$$= \frac{0.00475}{0.00475 + 0.00995}$$

$$= \frac{0.00475}{0.0147}$$

$$= 0.323$$

$$P(D/E) = 0.323$$

Therefore, if a person test result is positive then the probability of he (or) she having disease is

0.323.

Independent Events

From the conditional probability

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

Here The occurrence (or) non-occurrence of event F influence the occurrence of the event E .

If the occurrence (or) non-occurrence of the event F does not influence the occurrence of E then

$$P(E|F) = P(E).$$

$$\Leftrightarrow \frac{P(E \cap F)}{P(F)} = P(E)$$

$$\Leftrightarrow P(E \cap F) = P(E) P(F)$$

Two events E and F
are said to be independent
if $P(E \cap F) = P(E) \cdot P(F)$.

Eg: ① A card is selected
at random from a deck of
52 cards.

If E - event that the
selected card is an ace.

and

F - event that the selected card is a spade

then

$$P(E) = \frac{\binom{4}{1}}{52} = \frac{4}{52}$$

$$P(F) = \frac{\binom{13}{1}}{52} = \frac{13}{52}$$

$E \cap F$ - event that the selected card is an ace of spades

$$P(E \cap F) = \frac{1}{52}$$

$$P(E) \cdot P(F) = \frac{\cancel{4}}{\cancel{52}} \cdot \frac{\cancel{13}}{52}$$

$$= \frac{1}{52}$$

$$= P(E \cap F)$$

Events E and F are
independent.

Eg ②: Two coins flipped and

assume all 4 outcomes
are equally likely.

If E is the event that
the first coin lands on
heads and F is
the event that the
second coin lands on tails
then

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

$$E = \{ (H, H), (H, T) \}$$

$$F = \{ (H, T), (T, T) \}$$

$$E \cap F = \{ (H, T) \}$$

$$P(E \cap F) = \frac{1}{4}$$

$$P(E) = \frac{2}{4} ; P(F) = \frac{2}{4}$$

$$\boxed{P(E) P(F) = P(E \cap F)}$$

Thus E and F are independent.

Eg ③: Suppose that we toss two dice. Let

E_1 - event that sum of dice is 6

E_2 - event that sum of dice is 7

F - event that first die is 4

Then

$$E_1 = \{ (1,5), (2,4), (3,3), (4,2), (5,1) \}$$

$$E_2 = \{ (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) \}$$

$$F = \{ (4,1), (4,2), (4,3), (4,4), \dots \}$$

$$(4,5), (4,6) \}$$

$$P(E_1) = \frac{5}{36}, \quad P(E_2) = \frac{6}{36} = \frac{1}{6}$$

$$P(F) = \frac{6}{36} = \frac{1}{6}$$

$$P(E_1 \cap F) = \frac{1}{36} \quad \text{but} \quad P(E_1)P(F) = \frac{5}{216}$$

$$P(E_1 \cap F) \neq P(E_1) \cdot P(F)$$

E_1, F are not independent

However,

$$P(E_2 \cap F) = P(\{(4,3)\})$$

$$= \frac{1}{36}$$

$$P(E_2) \cdot P(F) = \frac{1}{6} \cdot \frac{1}{6}$$

$$= \frac{1}{36}$$

$$\boxed{P(E_2 \cap F) = P(E_2) \cdot P(F)}$$

E_2 and F are independent.

Result: If E is independent of F then E is independent of F^c .

Proof. Given that E is independent of F

$$p(E \cap F) = p(E) \cdot p(F).$$

Firstly, note that

$$E = (E \cap F) \cup (E \cap F^c)$$

It means that

$$\begin{aligned} p(E) &= p(E \cap F) + p(E \cap F^c) \\ &= p(E) \cdot p(F) + p(E \cap F^c) \end{aligned}$$

Equivalently,

$$\begin{aligned} p(E \cap F^c) &= p(E) - p(E) \cdot p(F) \\ &= p(E) [1 - p(F)] \\ &= p(E) p(F^c) \end{aligned}$$

Therefore, E is independent of F^c .

Remark: If E is independent of F and E is independent of G then E may not be independent of $F \cap G$.

Example: Suppose two dice are thrown. Let

E - event that sum of the dice is 7

F - event that the first die is 4

G - event that the second die is 3

then

$$E = \{ (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) \}$$

$$F = \{ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \}$$

$$G = \{ (1,3), (2,3), (3,3), (4,3), (5,3), (6,3) \}$$

$$P(E) = \frac{6}{36} = \frac{1}{6}$$

$$P(F) = \frac{6}{36} = \frac{1}{6}$$

$$P(G) = \frac{6}{36} = \frac{1}{6}$$

$$P(E \cap F) = \frac{1}{36} = P(E) \cdot P(F)$$

$$P(E \cap G) = \frac{1}{36} = P(E) \cdot P(G)$$

This shows that E is independent of F and E is independent of G .

$$\text{Since } P(E / F \cap G) = 1$$

$$\text{and } P(E) = \frac{1}{6}$$

we concluded that

E is not independent of $F \cap G$.

Problem: Suppose n independent trials is to be performed.

Each trial results in a success with probability ' p ' and failure with probability $1-p$. What is the probability that

(a) at least 1 success occurs in the n trials.

(b) exactly k success occurs in the n trials.

(c) All n trials result in success.

Solution:

Let E_i be the event that i^{th} trial results in failure.

$$P(E_i) = (1-p) \quad i=1, 2, \dots, n$$

(a) The probability of at least one success is,

$$P(E_1^c \cup E_2^c \cup E_3^c \cup \dots \cup E_n^c)$$

$$= P\left(\left(\bigcap_{i=1}^n E_i\right)^c\right)$$

$$= 1 - P\left(\bigcap_{i=1}^n E_i\right)$$

$$= 1 - P(E_1) P(E_2) \dots P(E_n)$$

$$= 1 - [(1-p)^n].$$

(b) Consider any particular k trials out of n trials.

The probability that these k -trials result in success and the rest $(n-k)$ fails is

$$p^k (1-p)^{n-k}.$$

Since there are $\binom{n}{k}$ ways of selecting k trials out of n trials, the desired probability is

$$P(\{\text{exactly } k \text{ successes}\})$$

$$= \binom{n}{k} p^k (1-p)^{n-k}$$

© The probability of all 'n' trials result in success is.

$$P(E_1^c \cap E_2^c \cap E_3^c \cap \dots \cap E_n^c)$$

$$= P(E_1^c) P(E_2^c) \dots P(E_n^c)$$

$$= p \cdot p \cdot p \dots p$$

$$= p^n.$$

Problem: Independent trials consisting of rolling a pair of dice are performed. What is the probability that an outcome of 5 appears before an outcome of 7? Here the outcome of a roll is the sum of dice.

Solution:

Let E_n denote the event that $(n \geq 1)$ neither 5 nor 7 appears as outcome on the first $(n-1)$ trials and 5 appears as outcome on the n^{th} trial

The desired probability is

$$P\left(\bigcup_{n=1}^{\infty} E_n\right).$$

Since $E_1, E_2, E_3, \dots$ are mutually exclusive events

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n).$$

The probability that 5 as outcome on any trial is

$$P\left(\{(1,4), (2,3), (3,2), (4,1)\}\right) \\ = \frac{4}{36} = \frac{1}{9}$$

The probability that 7 appears as outcome on any trial is,

$$P\left(\{ (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) \}\right)$$

$$= \frac{6}{36}$$

$$= \frac{1}{6}.$$

The probability that neither 5 nor 7 appears as outcome in any trial is,

$$= 1 - P(\text{either 5 or 7 as out come})$$

$$= 1 - P\left(\{(1,4), (1,6), (2,3), (2,5), (3,4), (3,2), (4,1), (4,3), (5,2), (6,1)\}\right)$$

$$= \left(1 - \frac{10}{36}\right)$$

Since E_n has first $(n-1)$ trials neither 5 nor 7 as out come and 5 appears on the n^{th} trials

$$P(E_n) = \left(1 - \frac{10}{36}\right)^{n-1} \frac{1}{9}$$

Thus

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

$$= \frac{1}{9} \sum_{n=1}^{\infty} \left(\frac{13}{18}\right)^{n-1}$$

$$= \frac{1}{9} \frac{1}{1 - \frac{13}{18}}$$

$$= \frac{1}{9} \times \frac{18^2}{5}$$

$$= \frac{2}{5}$$

The desired probability is $\frac{2}{5}$.



